Course: Cryptography and Network Security
Code: CSN15403
Branch: B.Tech -5th Semester

Lecture – 1: Introduction and Overview

Faculty & Coordinator : Dr. J Sathish Kumar (JSK)

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Prayagraj-211004

Cryptography & Network Security

Syllabus

Prerequisite: Computer Networks and Discrete Mathematics

L-T-P: 3-0-2, **Credits:** 4 **Type:** Core Essential Subjects (CES)

Course Description

In this course students will study the essential mathematical foundations for Information Security. This course features a rigorous introduction to modern cryptography, with an emphasis on the fundamental cryptographic primitives of public-key encryption, digital signatures, pseudo-random number generation, and basic protocols and their computational complexity requirements. After crediting these course students can look forward to wireless network security and E-commerce courses.

Course Objectives

Students undergoing this course are expected to:	
1	To understand basics of Cryptography and Network Security.
2	To be able to secure a message over insecure channel by various means
3	Learning the concepts of Message authentication and Hash functions.
4	To learn about how to maintain the Confidentiality, Integrity and Availability of a data
5	To understand various protocols for network security to protect against the threats in the networks.

Course Outcomes

CO Numbers	Course Outcomes (Action verb should be in italics)	Bloom's taxonomy
CO-1	**Identify** attacks types, security threats and their countermeasures	Identify/Knowledge
CO-2	**Understanding** and Symmetric and Asymmetric cryptography and basics of cryptanalysis	Understanding/ Comprehension:
CO-3	**Demonstrates and apply** cryptographic algorithms and protocols in attaining security in Network	Applying
CO-4	**Evaluate** security level and performance issues of different encryption and authentication techniques.	Analyzing
CO-5	**Develop** secure communication protocol	Creating

Syllabus

UNIT 1: Introduction to Cryptography, Block Cipher and Stream Cipher

Introduction to security attacks - services and mechanism - introduction to cryptography - Conventional Encryption: Conventional encryption model - classical encryption techniques - substitution ciphers and transposition ciphers – cryptanalysis – steganography - stream and block ciphers - Modern Block Ciphers: Block ciphers principals - Shannon's theory of confusion and diffusion - fiestal structure - data encryption standard(DES) - strength of DES - differential and linear crypt-analysis of DES - block cipher modes of operations - triple DES – AES.

UNIT II: Mathematical Background for Cryptography

Modulo Arithmetic, Introduction, Prime Number Generation, Shannon's Theory of Perfect Secrecy GCD Euclid's Algorithm, Algebraic Structures (Groups), Chinese Remainder Theorem.

Unit III: Public key cryptography and Authentication requirements) Principles of public key crypto systems - RSA algorithm - security of RSA - key management – Diffle-Hellman key functions - message authentication code - hash functions - birthday attacks – security of hash functions and MACS exchange algorithm - introductory idea of Elliptic curve cryptography – Elgamel encryption - Message Authentication and Hash Function: Authentication requirements - authentication

Syllabus

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UNIT IV (Integrity checks and Authentication algorithms): MD5 message digest algorithm - Secure hash algorithm (SHA) Digital Signatures: Digital Signatures - authentication protocols - digital signature standards (DSS) - proof of digital signature algorithm - Authentication Applications: Kerberos and X.509 - directory authentication service - electronic mail security-pretty good privacy (PGP) - S/MIME.

UNIT V (IPSecurity, Key Management, Web Security and System Security) IP Security: Architecture - Authentication header - Encapsulating security payloads - combining security associations - key management. Secure socket layer and transport layer security - secure electronic transaction (SET) - System Security: Intruders - Viruses and related threads - firewall design principals – trusted systems.

Text Books

1. Mao, Wenbo. Modern cryptography: theory and practice. Pearson Education India, 2003.
2. Schneier, Bruce. Applied cryptography: protocols, algorithms, and source code in C. john wiley& sons, 2007.
3. Stinson, Douglas R. Cryptography: theory and practice. Chapman and Hall/CRC, 2005.

Reference Books

1. Buchmann, Johannes. Introduction to cryptography. Vol. 335. New York: Springer, 2004.
2. Stallings, William. Cryptography and network security, 4/E. Pearson Education India, 2006.

Evaluation Process

(i) **Theory Type of Courses:** This type of course includes either lecture (at least 2 hours) or / and tutorial classes.

Mid-Sem Exam	Assessment	End-Sem Exam
25% (90 minutes duration)	25%	50% (150 minutes duration)

Assessment marks shall be awarded on the basis of Attendance, Surprise Tests, Quizzes, Assignments, Tutorials, etc.

Lets Get Started !!!!

History and Overview of Cryptography

- The Concise Oxford English Dictionary defines cryptography as "the art of writing or solving codes."
- Cryptography: The Science of creating coded messages
- Cryptanalysis: The art of breaking coded messages
- Clear text/Plaintext: The original Message
- Cipher text: The encoded message
- Key: Input to the cryptographic algorithm
- "Cryptography: Practice of the enciphering and deciphering of messages in secret code in order to render them unintelligible to all but the intended receiver

History and Overview of Cryptography

- But cryptography nowadays encompasses much more than this: it deals with
  - mechanisms for ensuring integrity,
  - techniques for exchanging secret keys,
  - protocols for authenticating users,
  - electronic auctions and elections,
  - digital cash, and more.
- Modern cryptography involves the study of mathematical techniques for securing digital information, systems, and distributed computations against adversarial attacks.

Early Cryptography

- 3500 BC: Sumerians: Cuneiform writings



Source: Internet

Early Cryptography

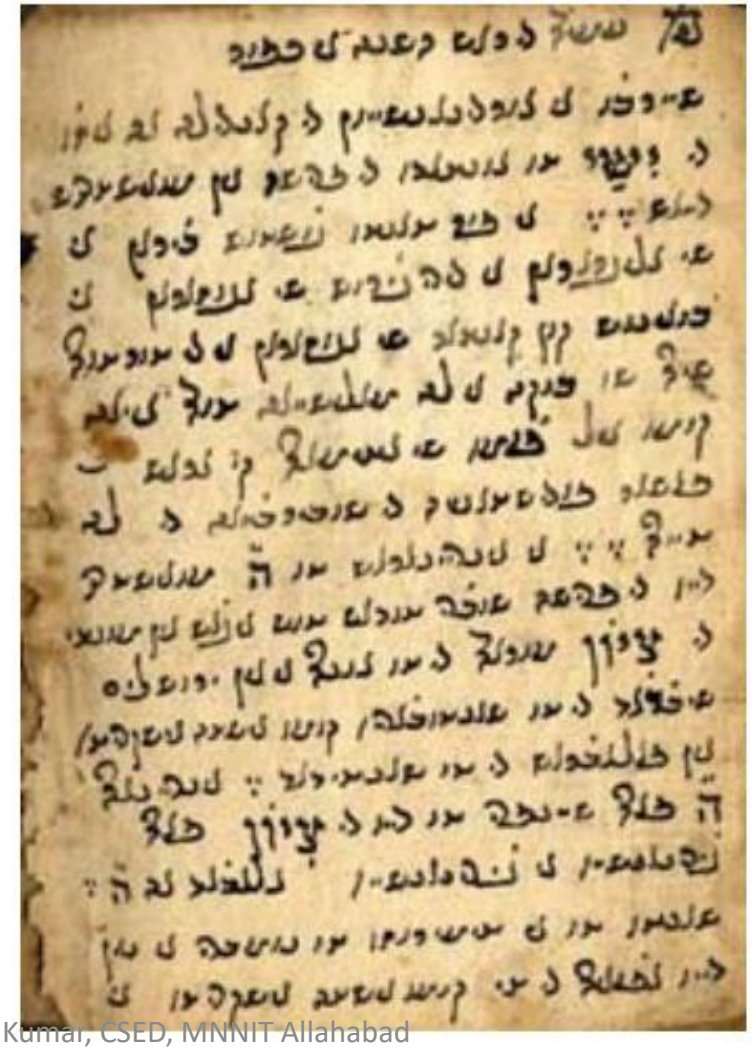
- 1900 BC: Egypt: First known use of cryptography



Source: Internet

Early Cryptography

- 500 – 600 BC: ATBASH Cipher
- Used by Hebrew scribes – Substitution cipher (reversed alphabet)



Source: Internet

Early Cryptography

- 486 BC: Greece
- σκυτάλη – skytale



Source: Internet

Early Cryptography

- 60 – 50 BC: Julius Caesar – substitution cipher – Shift letters by X positions:
- E.g. X = 3: A -> D, B -> E, C -> F,....
- Weakness?
  - Frequency analysis (1000 AD)



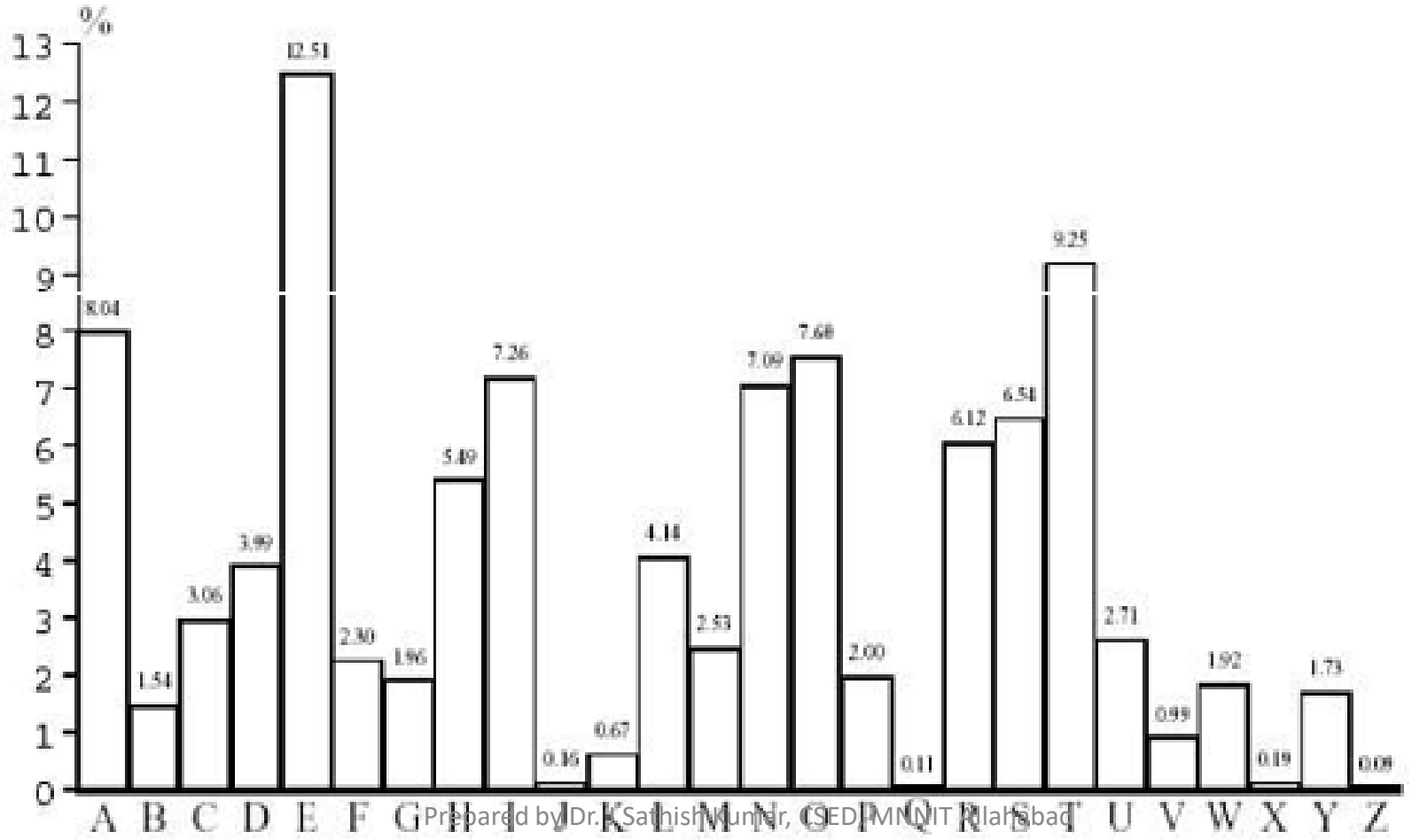
Source: Internet

Early Cryptography

- Mono-alphabetic substitution
  - letters of the plain text alphabet are mapped on to unique letters throughout the entire message text
  - cipher can be trivially broken because

    i. The language of the plain text is easily recognizable. (frequency distribution- unigram statistics- next slide)
    ii. There are only s= |A| keys (e.g. for Roman alphabet, only 25 keys – 1 to 25) to search exhaustively
    - Exhaustive key search is always possible <make it practically infeasible is the goal>

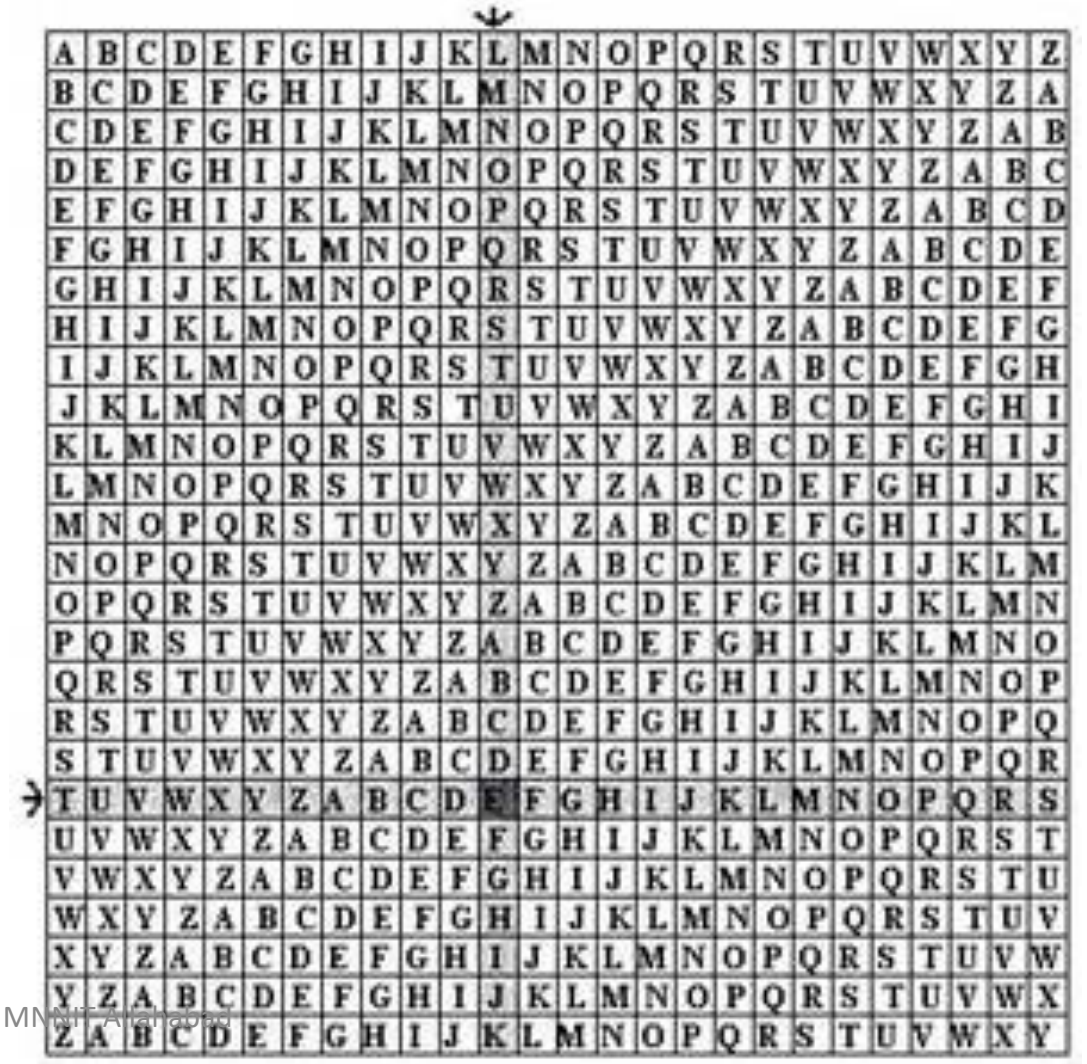
Frequency of single characters in English text



Medieval Cryptography

- 1587: Vigenère Cipher
- Polyalphabetic: one to many relationship
- Example

plaintext:	T	O	B	E	O	R	N	O	T	T	O	B	E
key:	L	T	O	B	E	O	R	N	O	T	T	O	B
ciphertext:	E	H	P	F	S	F	E	B	H	M	H	P	F



Medieval Cryptography
Vigenère Cipher

Example:
Message: "What fools these mortals be"

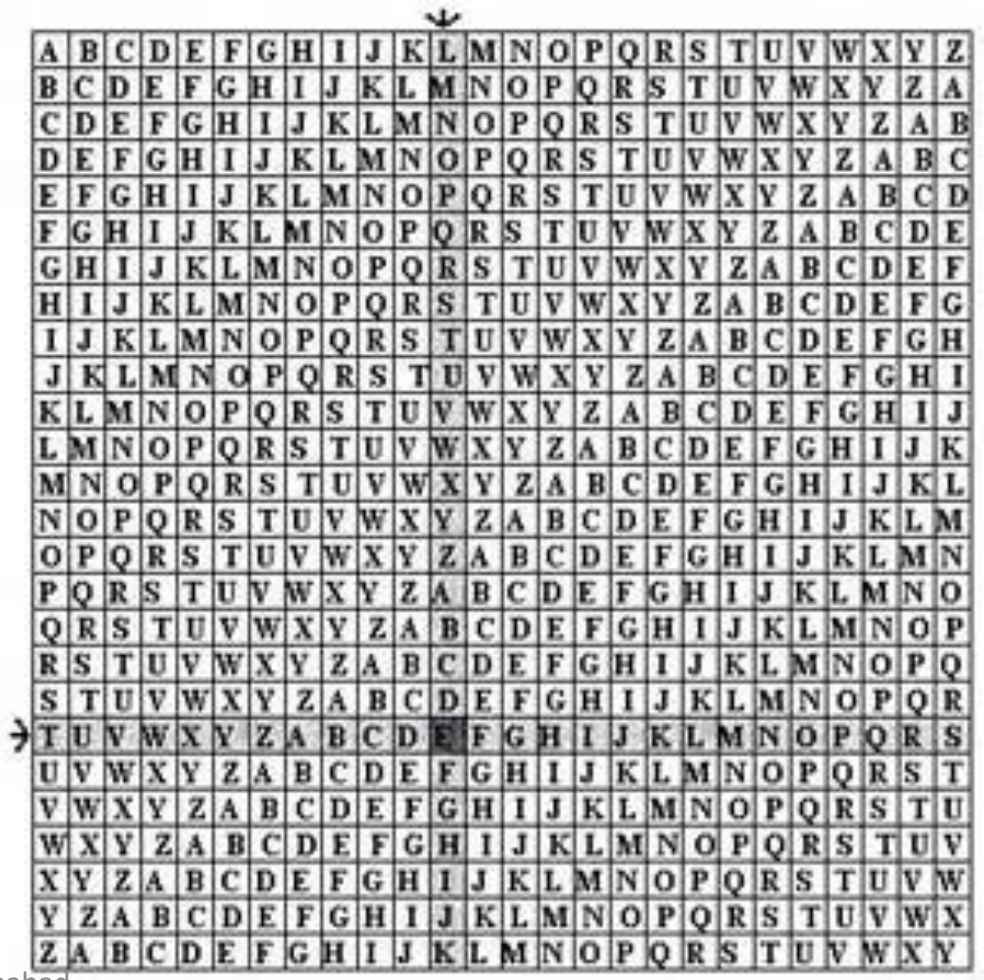
Keyword: Puck

plaintext:	W	H	A	T	F		O	O	L	S	T		H	E	S	E	M		O	R	T	A	L		S	B	E
key:	P	U	C	K	P		U	C	K	P	U		C	K	P	U	C		K	P	U	C	K		P	U	C
ciphertext:	L	B	C	D	U		I	Q	V	H	N		J	O	H	Y	O		Y	G	N	C	V		H	V	G

Example:
Message: LBCDU IQVHN JOHYO YGNCV HVG

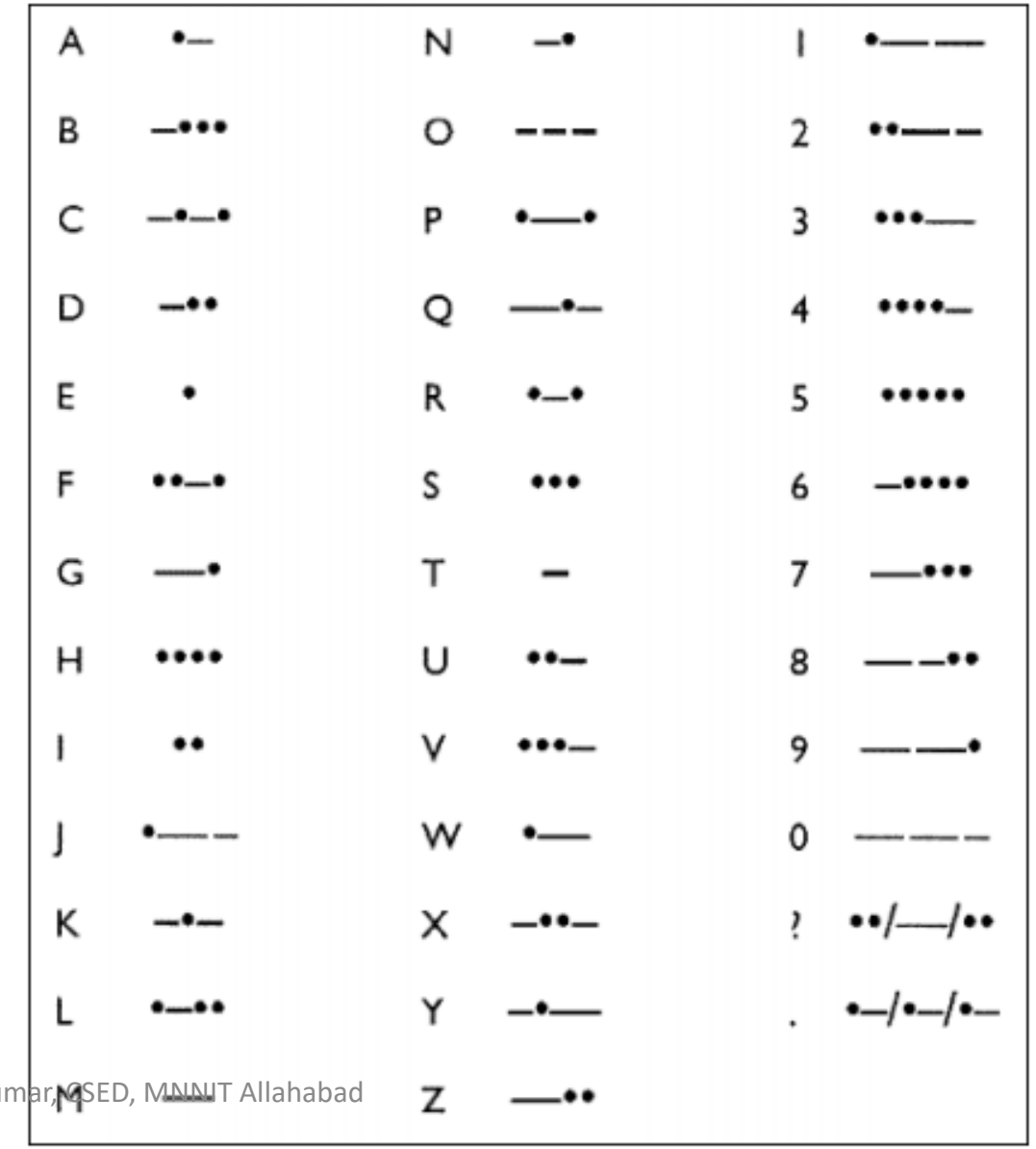
Keyword: Puck

key:	P	U	C	K	P		U	C	K	P	U		C	K	P	U	C		K	P	U	C	K		P	U	C
ciphertext:	L	B	C	D	U		I	Q	V	H	N		J	O	H	Y	O		Y	G	N	C	V		H	V	G
plaintext:	W	H	A	T	F		O	O	L	S	T		H	E	S	E	M		O	R	T	A	L		S	B	E



Modern Cryptography

- 1845: Morse Code –
- Representation by code signal
- States (on and off) composed into 5 symbols



A	•–	N	–•	1	•——
B	–•••	O	–––	2	••—–
C	–•–•	P	•––•	3	•••—–
D	–••	Q	––•–	4	••••–
E	•	R	•–•	5	•••••
F	••–•	S	•••	6	–••••
G	––•	T	–	7	––•••
H	••••	U	••–	8	––••
I	••	V	•••–	9	––––•
J	•–––	W	•––	0	–––––
K	–•–	X	–••–	?	••/––/••
L	•–••	Y	–•––	.	•–/•–/•–
M	––	Z	––••		

Modern Cryptography

- 1863: Friedrich Kasiski breaks Vigenere: –
  - By examining repeated strings of characters in the cipher text, which could indicate the length of the secret key.
  - Find length of keyword
  - Once the length of the secret key is known, the cipher text is rewritten into a corresponding number of columns, with a column for each letter of the key.
  - Each column is then made up of plaintext that's been encrypted by one Caesar cipher.
  - Use frequency analysis to solve these

Modern Cryptography

- 1918: ADFGVX Cipher – Used in the German army in WWI

Encrypt the plaintext "attack at 1200am"
using keywords *147 regiment* and *privacy*

	A	D	F	G	V	X
A	1	4	7	R	E	G
D	I	M	N	T	A	B
F	C	D	F	H	J	K
G	L	O	P	Q	S	U
V	V	W	X	Y	Z	0
X	2	3	5	6	8	9

a	t	t	a	c	k	a	t	1	2	0	0	a	m
DV	DG	DG	DV	FA	FX	DV	DG	AA	XA	VX	VX	DV	DD

P	R	I	V	A	C	Y
4	5	3	6	1	2	7
D	V	D	G	D	G	D
V	F	A	F	X	D	V
D	G	A	A	X	A	V
X	V	X	D	V	D	D

Cipher Text: "DXXV GDAD DAAX DVDX VFGV GFAD DVVD".

Modern Cryptography

- 1918: ADFGVX Cipher – Used in the German army in WWI

Decrypt the ciphertext "ADDDF DDAXF XAGGF DXXAX FGXFG G" which was encrypted using keywords *monkeys* and *zebras*

By reading off each row we get the intermediate text "GXFGAX XFDFDA FXDDDX GAAXDF GG".

Z	E	B	R	A	S
6	3	2	4	1	5
G	X	F	G	A	X
X	F	D	F	D	A
F	X	D	D	D	X
G	A	A	X	D	F
G	G				

	A	D	F	G	X
A	M	O	N	K	E
D	Y	S	A	B	C
F	D	F	G	H	I
G	L	P	Q	R	T
X	U	V	W	X	Z

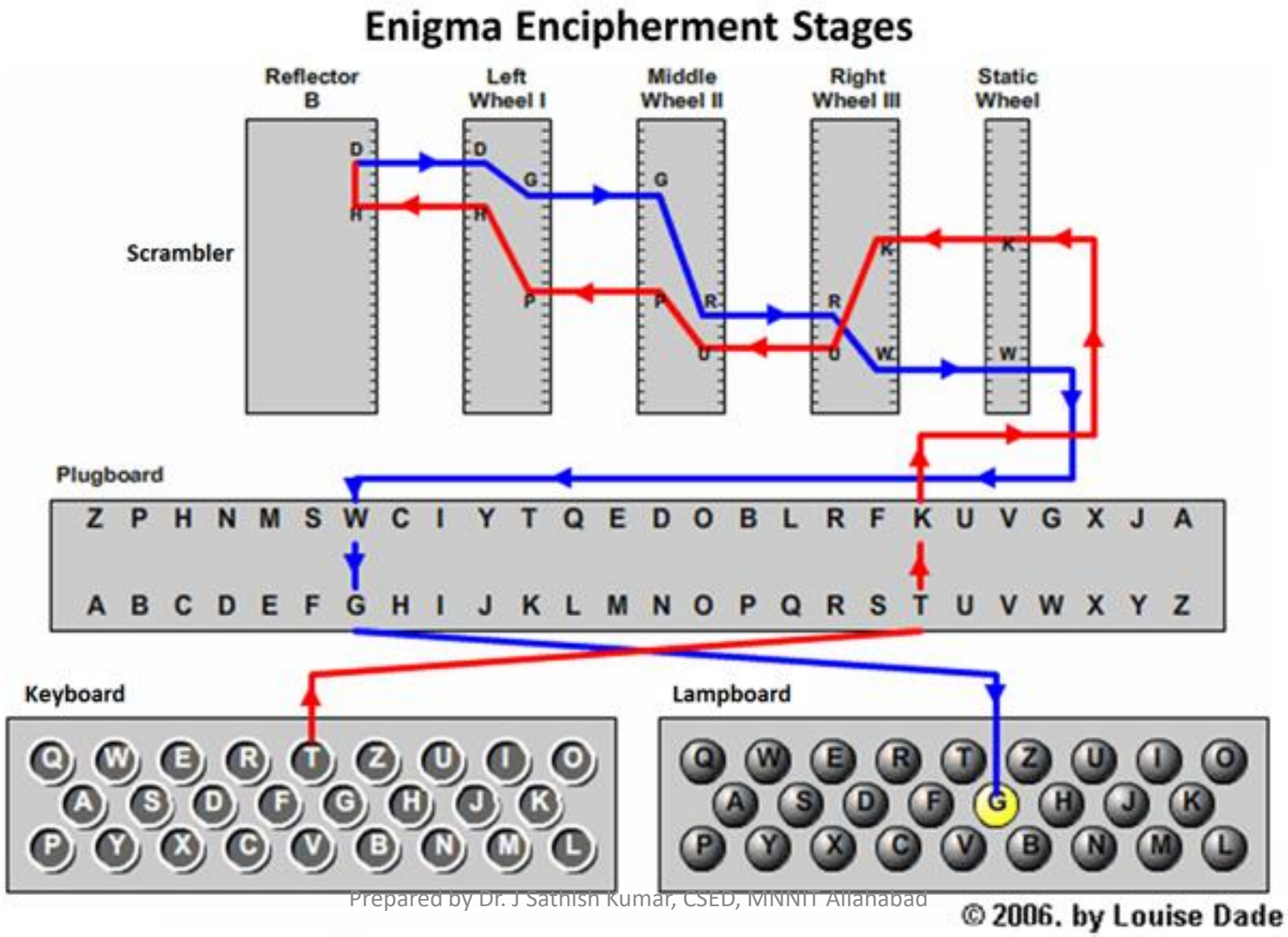
The plaintext is then retrieved as "the way is clear".

Modern Cryptography

- 1918: The Enigma – Arthur Scherbius
- Business: confidential docs
- No codebooks
- Rotors -> multi substitution
- Wireing changes as-youtype
- German forces in WWII

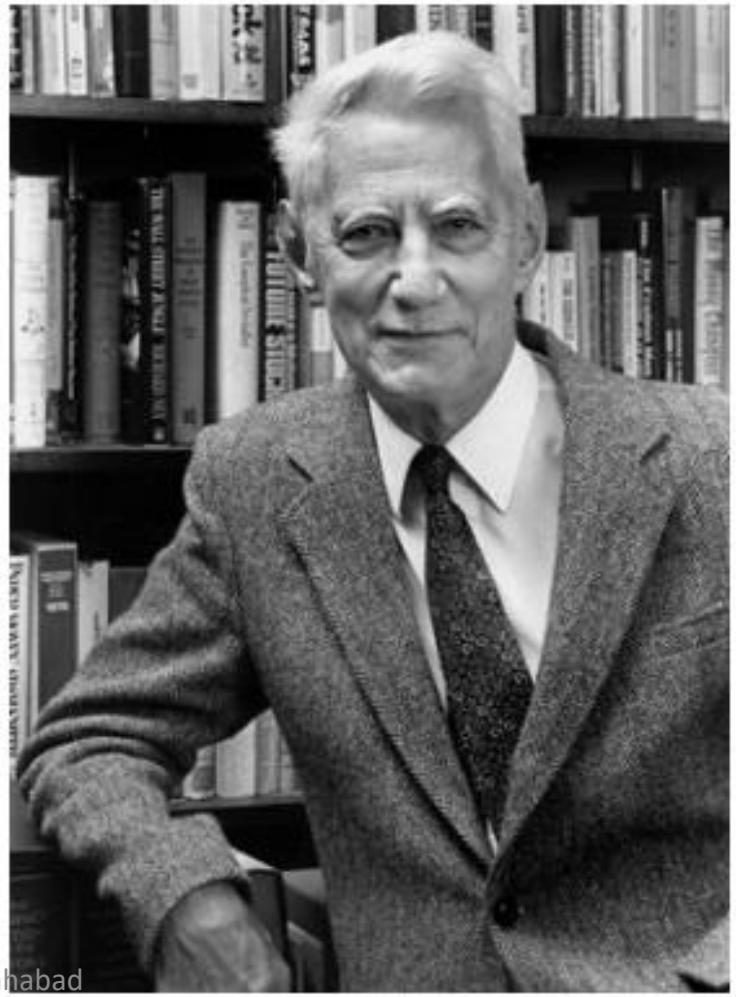


Modern Cryptography: The Enigma



Modern Cryptography

- 1949: Shannon: –
- Communication Theory of Secret Systems
- Proved: One time pad unbreakable



Modern Cryptography

- The DES (Data Encryption Standard) algorithm is a symmetric-key block cipher created in the early 1970s by an IBM team and adopted by the National Institute of Standards and Technology (NIST).
- 1976: Diffie – Hellman Key Exchange
- Public Key Crypto –
  - Key exchange problem –
- Asymmetric key algorithm – E.g: RSA, MIT, 1977
  - The most common public-key algorithm is the RSA cryptosystem, named for its inventors (Rivest, Shamir, and Adleman).
- Elliptic Curve Cryptography (ECC) was discovered in 1985 by Victor Miller (IBM) and Neil Koblitz (University of Washington) as an alternative mechanism for implementing public-key cryptography.
- MD5 - Ronald Rivest, founder of RSA Data Security and professor in MIT, 1991
- The Advanced Encryption Standard (AES) was published on November 26, 2001
- SHA-512-2001

Join in Google Class ID

- Class Name and Code For Section C Students:
  - CS5C - Cryptography and Network Security (CSN15403)
  - Class Code for Joining: CNS5C-2025 for Section C
- Class Name and Code For Section D Students:
  - CS5D - Cryptography and Network Security (CSN15403)
  - Class Code for Joining: CNS5D-2025 for Section D

- Purpose:
  - Assignment Submissions
  - PPTs and Materials Accessibility
  - Feedback

Assignment #0

- Watch Movie “The Imitation Game”